

i.e. $\Delta = \{\delta : \delta_t = \gamma \cdot t, \gamma \in \mathbb{R}\}$. There are often concerns, however, that the linear specification is not exactly correct (Wolfers, 2006; Lee and Solon, 2011). A natural relaxation is therefore to impose only that the differential trends evolve smoothly over time by bounding the extent to which its slope may change across consecutive periods. Such a restriction can be formalized in our framework by imposing that $\delta \in \Delta^{\text{SD}}(M)$ for $M \geq 0$, where

$$\Delta^{\text{SD}}(M) := \{\delta : |(\delta_{t+1} - \delta_t) - (\delta_t - \delta_{t-1})| \leq M, \forall t\}. \quad (8)$$

The parameter $M \geq 0$ governs the amount by which the slope of δ can change between consecutive periods, and thus bounds the discrete analogue of the second derivative. We use the abbreviation SD for “second differences” or “second derivative.”⁹ In the special case where $M = 0$, $\Delta^{\text{SD}}(0)$ requires that the difference in trends be exactly linear, which corresponds with the assumption underlying the parametric linear specification common in applied work.

2.4.4. Example: non-staggered DiD (continued). In the three-period DiD model, assuming the differential trend is exactly linear is equivalent to assuming $\Delta = \{\delta : \delta_1 = -\delta_{-1}\}$. Assuming $\delta \in \Delta^{\text{SD}}(M)$ requires only that the linear extrapolation be *approximately* correct, $\delta_1 \in [-\delta_{-1} - M, -\delta_{-1} + M]$.

2.4.5. Combining smoothness and relative magnitudes bounds. In some contexts, researchers may be willing to assume that the difference in trends evolves relatively smoothly over time but may be unsure about the smoothness bound $M \geq 0$ introduced above. In such cases, it may be reasonable to assume that the possible non-linearities in the post-treatment difference in trends are bounded by the observed non-linearities in the pre-treatment difference in trends. This can be formalized with the restriction

$$\Delta^{\text{SDRM}}(\bar{M}) = \{\delta : \forall t \geq 0, |(\delta_{t+1} - \delta_t) - (\delta_t - \delta_{t-1})| \leq \bar{M} \cdot \max_{s < 0} |(\delta_{s+1} - \delta_s) - (\delta_s - \delta_{s-1})|\},$$

which bounds the maximum deviation from a linear trend in the post-treatment period by $\bar{M} \geq 0$ times the equivalent maximum in the pre-treatment period. The set $\Delta^{\text{SDRM}}(\bar{M})$ is thus similar to $\Delta^{\text{SD}}(M)$ introduced above, except it allows the magnitude of the possible non-linearity to explicitly depend on the observed pre-trends.

2.4.6. Sign and monotonicity restrictions. Context-specific knowledge may sometimes also suggest sign or monotonicity restrictions on the differential trend. For instance, if the policy of interest occurs at the same time as a confounding policy change that we expect to have a positive effect on the outcome, we might restrict the post-treatment bias to be positive, $\delta \in \Delta^{\text{PB}} := \{\delta : \delta_t \geq 0 \forall t \geq 0\}$. Likewise, there may be secular pre-existing trends that we expect would have continued following the treatment date.¹⁰ We may then wish to impose that the differential trend be increasing, $\delta \in \Delta^{\text{I}} := \{\delta : \delta_t \geq \delta_{t-1} \forall t\}$, or monotone with unknown sign, $\delta \in \Delta^{\text{Mon}} := \Delta^{\text{I}} \cup (-\Delta^{\text{I}})$. Sign and monotonicity restrictions may be combined with the

9. Restrictions on the second derivative of the conditional expectation function or density have been used in regression discontinuity settings (Kolesár and Rothe, 2018; Frandsen, 2016; Noack and Rothe, 2020). Smoothness restrictions are also used to obtain partial identification in e.g. Manski (1997); Kim *et al.* (2018).

10. Monotone violations of parallel trends are often discussed in applied work. For example, Lovenheim and Willen (2019a) argue that violations of parallel trends cannot explain their results because “pre-[treatment] trends are either zero or in the wrong direction (i.e. opposite to the direction of the treatment effect).” Greenstone and Hanna (2014) estimate upward-sloping pre-existing trends and argue that “if the pre-trends had continued” their estimates would be upward biased.